A CURATED TECHNICAL ARCHIVE

AWS and Cloud Computing

AWS Certified Cloud Practitioner, CLF-C01

Cloud Computing Laboratory, University of Mumbai

21	9	12
MODULES	EXPERIMENTS	VIDEO LESSONS

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A Note from Amey Thakur

Education is not a destination but a continuous journey. This archive represents my commitment to that philosophy: a deliberate effort to preserve, organise, and revisit the cloud computing knowledge acquired while preparing for the AWS Certified Cloud Practitioner examination and completing the Cloud Computing Laboratory.

By making this work publicly available, I hope it serves not only my own continued education but also assists others on their learning journeys.

> **Important**
> Cloud platforms change quickly. What does not go stale is the shape of the subject: what a region is, why the shared responsibility model is drawn where it is, what a service actually costs.

What is in here

Read it	Eleven Academy modules and ten sets of written notes, covering the same syllabus twice.
Build it	Nine laboratory experiments, each with a written report.
Watch it	Twelve recordings of the console being driven, not described.
Check it	Assignments, quizzes and reference material, kept as they were assessed.

> **Note**
> I compiled this while preparing for the AWS Certified Cloud Practitioner examination and completing the Cloud Computing Laboratory in the final semester of my B.E.

— Amey Thakur

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Start Here

What is in here.

21	9	12	2h 34m
MODULES	EXPERIMENTS	RECORDINGS	OF VIDEO

Who it is for

You are sitting CLF-C01	Ten sets of written notes and eleven course modules covering the syllabus, plus the assignments and quizzes as they were assessed.
You want the hands-on part	Nine laboratory experiments with full reports, and recordings of the console being driven through most of them.
You are teaching this	Everything is CC BY 4.0, so you can take it into your own material.

Tip
Everything linked from this document is published under Creative Commons Attribution 4.0. You may read it, copy it, adapt it and build on it, including commercially. Credit me and you are done.

Four ways in

Studying for CLF-C01
Start with the written notes. Ten modules covering the certification syllabus in prose.

Written Notes by Topic

Here to build something
Nine laboratory experiments, each with a report showing what was done and what it cost.

Cloud Computing Laboratory

Would rather watch
Twelve recordings of the AWS console being driven through the work.

Video lessons

Want to see it running
One experiment deploys to a site that is still live today.

amey-thakur.github.io/CLOUD-COMPUTING-LAB

How to Use This Archive

Start with the learning path	Eleven Academy modules, in order, each with notes on the same topic.
Read, then build	Every experiment has a report. Most have a recording, so you can watch the console before you drive it.
The repository is the material	This document indexes; it does not duplicate. Every entry links to the file.
Verify pricing against AWS	Costs, limits and console layouts change. What is stated here was true when the work was done.

> **Important**
> This is my own resource. It is not produced, sponsored, endorsed or certified by Amazon Web Services, and it is not official AWS training material. AWS, Amazon Web Services and the AWS service names are trademarks of Amazon.com, Inc. or its affiliates, used here only to identify the subject.

The Learning Path

The AWS Academy Cloud Foundations course runs in eleven modules, from cloud concepts through to automatic scaling and monitoring. The archive holds the module material and a parallel set of written notes covering the same ten topics, so each subject can be read twice, once as slides and once as prose.

Topic	Course module	Written notes
Course Introduction	Module 00	—
Cloud Concepts	Module 01	MODULE 1
Cloud Economics and Billing	Module 02	MODULE 1
AWS Global Infrastructure	Module 03	MODULE 3
AWS Cloud Security	Module 04	MODULE 1
Networking and Content Delivery	Module 05	MODULE 5
Compute Services	Module 06	MODULE 6
Storage	Module 07	MODULE 7
Databases	Module 08	MODULE 8
Cloud Architecture	Module 09	MODULE 1
Automatic Scaling and Monitoring	Module 10	MODULE 10

Written Notes by Topic

Ten modules of written notes, each covering one area of the certification syllabus. These are the reading half of the archive.

Module	Topics covered
MODULE 1: Cloud Concepts	Cloud Computing Overview, Service Models, Deployment Models
MODULE 2: Billing and Economy	AWS Billing, Pricing Models, TCO, Cost Optimization
MODULE 3: AWS Global Infrastructure	Regions, Availability Zones, Edge Locations
MODULE 4: AWS Cloud Security	IAM, Shared Responsibility, Security Groups
MODULE 5: Networking and Content Delivery	VPC, Subnets, CloudFront, Route 53
MODULE 6: Compute Services Overview	EC2, Lambda, Elastic Beanstalk, Lightsail
MODULE 7: Storage	S3, EBS, EFS, Storage Gateway
MODULE 8: Databases	RDS, DynamoDB, Aurora, Redshift
MODULE 9: Cloud Architecture	Well-Architected Framework, Design Principles
MODULE 10: Automatic Scaling and Monitoring	Auto Scaling, CloudWatch, Load Balancing

Cloud Computing Laboratory

Nine experiments from the Cloud Computing Laboratory, taken in the final semester of the B.E. at the University of Mumbai. Each has a written report. Six carry a recorded demonstration, and one deploys to a live site that is still running.

Experiment 1. Study of NIST model of cloud computing

Completed January 10, 2022 · assessed 10/10

Report

Experiment 2A. Implement Software Virtualization using Hypervisors (VMware)

Completed January 17, 2022 · assessed 10/10

Report

Experiment 2B. Implement Virtualization using VirtualBox

Completed January 24, 2022 · assessed 10/10

Report

Experiment 3. Demonstrate and implement IaaS service using AWS EC2

Completed January 31, 2022 · assessed 10/10

Report · Video lesson

Experiment 4. Demonstrate and implement Storage as a service using AWS S3

Completed February 14, 2022 · assessed 10/10

Report · Video lesson

Experiment 5. Demonstrate and implement PaaS – Deploy static website using AWS S3

Completed February 21, 2022 · assessed 10/10

Report · Video lesson · Live deployment

Experiment 6. Deploy WordPress app using Lightsail service in AWS (SaaS)

Completed February 23, 2022 · assessed 10/10

Report · Video lesson

Experiment 7. Understand Security of Web Server and demonstration of IAM using AWS

Completed February 23, 2022 · assessed 10/10

Report · Video lesson

Experiment 8. Understanding implementation and applications of Fog Computing

Completed March 10, 2022 · assessed 10/10

Report

Original Video Lessons

I recorded these while doing the laboratory work. Each shows the AWS console being driven through the experiment it goes with.

> **Tip**
> Every thumbnail below is a link. 12 videos, 2 hours 34 minutes in total, on my own channel.

Full playlist	youtube.com/playlist?list=PLGOc13Pt03SZuZA2eS79J4EUtBBgROJCs
Channel	youtube.com/@ameythakur

Amazon EC2 Instance Tutorial
6:07

Amazon S3 Tutorial
10:11

Amazon EBS Volume Tutorial
7:20

Amazon EFS Tutorial
9:45

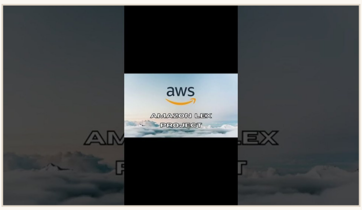
Amazon RDS Tutorial
9:49

Amazon VPC AuroraDB Tutorial
25:18

Amazon Load Balancer Tutorial
28:04

Amazon Lightsail WordPress Tutorial
8:08

Amazon IAM Tutorial
4:27



Pizza Ordering Chatbot using Amazon Lex - AWS Project Demo
1:53

Pizza Ordering Chatbot using Amazon Lex - AWS Project
42:15

Pizza Ordering Chatbot using Amazon Lex - Presentation
1:11

Recordings mapped to experiments

Experiment	What the recording shows	Watch
Experiment 3	Demonstrate and implement IaaS service using AWS EC2	Video
Experiment 4	Demonstrate and implement Storage as a service using AWS S3	Video
Experiment 5	Demonstrate and implement PaaS - Deploy static website using AWS S3	Video
Experiment 6	Deploy WordPress app using Lightsail service in AWS (SaaS)	Video
Experiment 7	Understand Security of Web Server and demonstration of IAM using AWS	Video

Practice and Assessment

The assessed work from the laboratory, kept because a worked assignment shows what the questions actually look like.

Assignments

Item	Topics	Date
Assignment 1	SOA, Google App Engine, Cloud Service Models, Hypervisor, Virtualization, OpenStack	February 15, 2022
Assignment 2	Cloud Security, IAM, IaaS, PaaS, SaaS Service Models	February 15, 2022

Quizzes

Item	Topics	Date
Quiz 1	Introduction to Cloud Computing	January 27, 2022
Quiz 2	Practice Quiz	February 17, 2022

Reference Material

Books, lecture material and official AWS learning platforms. The certification guide and the AWS Ramp-Up Guide are the two worth reading first.

Resource	What it is
AWS Certified Cloud Practitioner (CLF-C01) Cert Guide	Core Reference Guide (Anthony Sequeira)
AWS Ramp-Up Guide Cloud Foundations	Official AWS Learning Roadmap
CCL Presentation	Cloud Computing Lab overview slides
CCL Oral Exam QB	Oral/Viva preparation questions
CCS Easy Solution	Comprehensive solved questions
Characteristics	Cloud characteristics and models
Cloud Computing PDF	Core cloud computing concepts
Cloud Computing	Detailed cloud computing slides
CCL Notes	Comprehensive subject notes
NPTEL Lecture Material	NPTEL video lecture materials

Official AWS resources

AWS Skill Builder	Amazon's own learning platform. Official source
AWS Documentation	The authority on current service behaviour, limits and pricing. Official source

Repository Directory

Repository	What it holds
AWS-CERTIFIED-CLOUD-PRACTITIONER-CLF-C01	Course modules, written notes, reference books, staged laboratory work, assessments and a mini-project
CLOUD-COMPUTING-LAB	Nine laboratory experiments with reports, assignments, quizzes, reference books and the Pizza Ordering Chatbot built on Amazon Lex
Live deployment	The static site deployed during Experiment 5, still running

Elsewhere

GitHub	github.com/Amey-Thakur
Amey's Arc	amey-thakur.github.io
ORCID	0000-0001-5644-1575
Google Scholar	Publications and citations
ResearchGate	researchgate.net/profile/Amey-Thakur
arXiv	arxiv.org/a/thakur_a_3
Kaggle	kaggle.com/ameythakur20
Hugging Face	huggingface.co/ameythakur
LinkedIn	linkedin.com/in/amey-thakur
YouTube	Engineering project demonstrations

What I Would Tell You

Six things I learned building this, offered for whatever they are worth.

1 **Do the laboratory, do not read it.**

Cloud concepts are cheap to read and expensive to actually understand. Nothing taught me what a security group is like being locked out by one.

2 **Turn everything off.**

I ended every experiment by tearing the stack down. Free tier is not free once you leave a load balancer running, and the habit is worth more than the money.

3 **Screenshot the console as you go.**

Every report here has them. The console layout has already changed since; the screenshots are what makes the reports still readable.

4 **Learn the shared responsibility model before anything else.**

Most of the certification, and most real mistakes, come back to which side of that line a thing sits on.

5 **Record the walkthrough while it still works.**

Twelve of these recordings exist because I made them on the day. Some of those stacks could not be rebuilt now.

6 **Check pricing against AWS, not against me.**

Every figure here was true when the work was done. Treat it as a shape, not a quote.

> **Tip**
> The certification is the smallest thing you get out of this. The habit of building it, breaking it and tearing it down is the rest.

A Closing Note

The repositories are the material. This document is an index to them, written so that someone arriving without context can find the part they need.

> **Important**
> Cloud services change faster than anything else here. Where I state a price, a limit or a console step, it was true when the work was done. AWS documentation is the authority now.

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